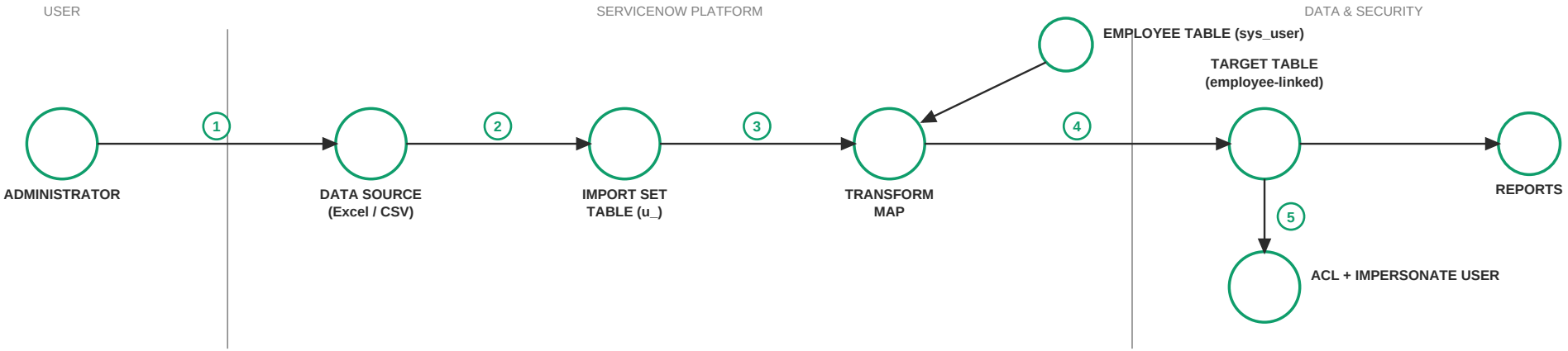
Project Design Phase-II
Data Flow Diagram & User Stories

Date	03 October 2026
Team ID	
Project Name	Employee Data Import using Transform Maps (Spreadsheet) - ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

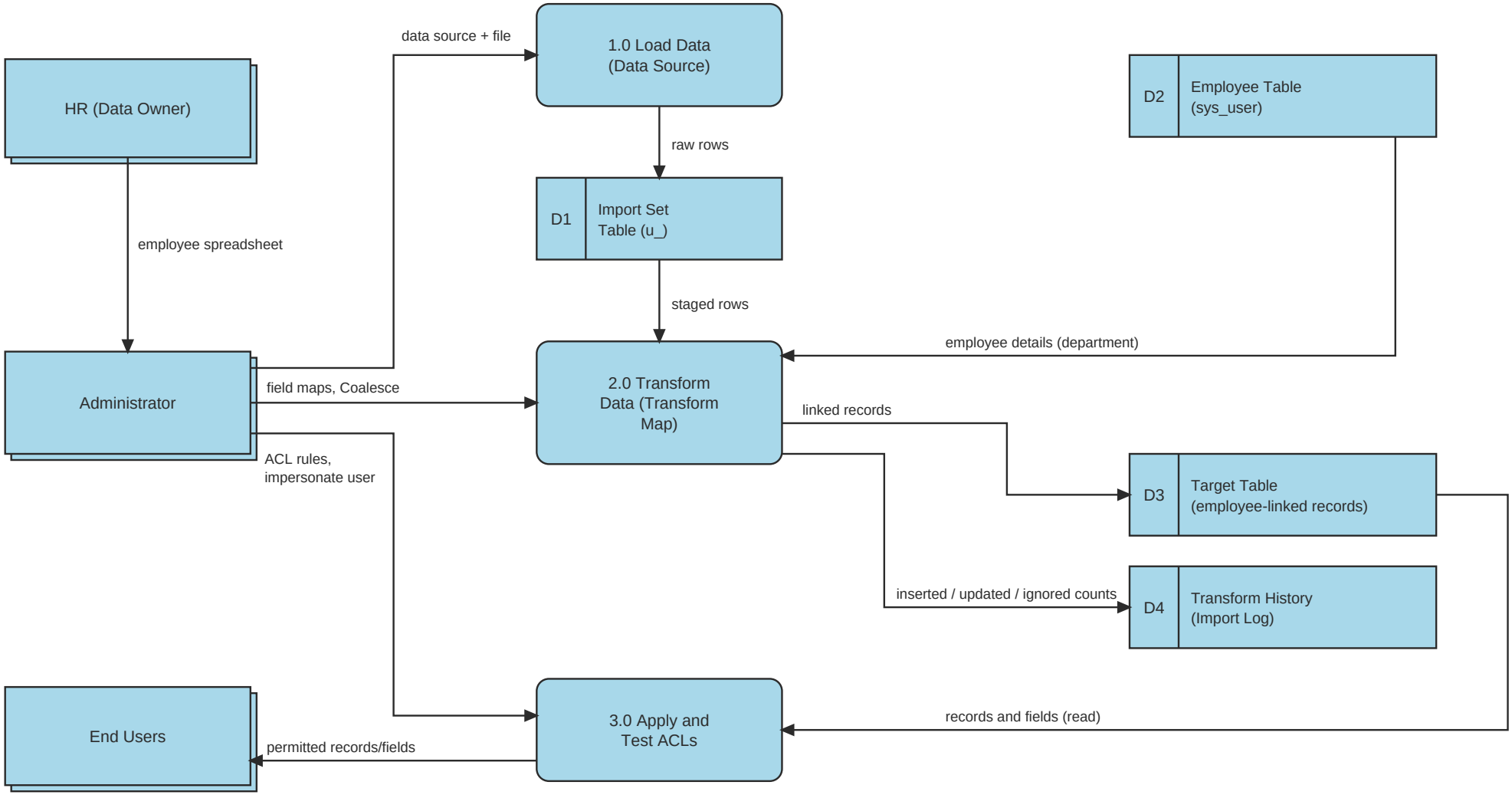
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored. For this project, it shows how employee data moves from a spreadsheet through a Data Source, staging table and Transform Map into employee-linked records that are protected by ACLs.

Example: (Simplified)



1. Administrator receives the employee spreadsheet (Excel/CSV) and attaches it to a Data Source.
2. Administrator loads the file; the rows are copied into the Import Set (staging) table.
3. The Transform Map maps staging fields to the target table; Employee ID is linked to the Employee (sys_user) record and Coalesce prevents duplicates.
4. Department and other employee details are pulled into each record and the results are logged in Transform History.
5. ACLs are created on the target table and fields, and tested using Impersonate User; the secured data is then used in reports.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Data Import	USN-1	As an administrator, I can create a Data Source and attach the employee spreadsheet (Excel/CSV).	File is attached and the format and header settings are saved	High	Sprint-1
		USN-2	As an administrator, I can load the spreadsheet rows into a staging (Import Set) table.	All rows are visible in the Import Set table	High	Sprint-1
	Transform Map	USN-3	As an administrator, I can create a Transform Map from the staging table to the target table.	Transform Map is saved with the correct source and target	High	Sprint-1
		USN-4	As an administrator, I can map spreadsheet fields, including Employee ID to the Employee (sys_user) reference field.	Field Maps list shows all mapped columns and each record is linked to an employee	High	Sprint-1
		USN-5	As an administrator, I can set Coalesce on Employee ID so existing records are updated and not duplicated.	Re-import shows updated records and no duplicates	Medium	Sprint-2
		USN-6	As an administrator, I can pull the employee's department into the record automatically.	Department is auto-filled and visible in the list view	High	Sprint-2
	Verification	USN-7	As an administrator, I can review Transform History and the Import Log after a run.	Inserted, updated and ignored counts are shown and errors can be identified	Medium	Sprint-2
	Security (ACL)	USN-8	As an administrator, I can create ACLs on the target table and its fields.	ACL records are created with the required roles	High	Sprint-3
		USN-9	As an administrator, I can test access using Impersonate User.	Impersonated user sees only the permitted records and fields	High	Sprint-3
HR Data Owner / Manager	Reporting	USN-10	As an HR user, I can view employee-linked records with department details and report by department.	Report shows record counts grouped by department	Medium	Sprint-3
Employee (End User)	Access Control	USN-11	As an employee, I can view only the records and fields I am permitted to see.	Other employees' sensitive fields are hidden	High	Sprint-3